

Modelling Individual Skill in Collaborative Groups - Attribution-based Bayesian Knowledge Tracing

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Abstract

When learners work together to produce a shared solution, the resulting outcome does not necessarily show what each learner understands as different learners may contribute differently to the same solution. We introduce Attribution-based Bayesian Knowledge Tracing (A-BKT), which maintains separate estimates of individual skill knowledge from shared collaborative observations. A-BKT adjusts the influence of each observation according to the learner's estimated contribution to the evolving solution. We use process information from intermediate task states to estimate these contributions and examine whether separating evidence collected during the task from evidence collected at task completion improves individual knowledge estimates. We evaluate A-BKT using data from 13 adult dyads completing collaborative Snap Circuits tasks. Individual mastery estimates were assessed against post-test performance and video annotations of learner-specific evidence during the activity. The resulting estimates showed limited ability to distinguish higher from lower post-test performance. However, learner-specific attribution improved the model's ability to distinguish between partners, while the proposed evidence-handling approach improved correspondence with the video annotations. Overall, the findings show that shared collaborative observations can support individual knowledge traces when combined with information about learner contribution.

Keywords

Skill Modelling, Knowledge Tracing, Tutoring, Group Learning, Collaborative Learning

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1. Introduction

Knowledge tracing estimates how a learner's knowledge of particular skills changes across repeated learning opportunities (Abdelrahman et al., 2023; Corbett & Anderson, 1994). These estimates can support adaptive learning systems by helping them decide when a learner may need additional practice, explanation, or feedback (Evans et al., 2019). Most knowledge-tracing approaches assume that the observed response used to update a learner's knowledge estimate can be associated with that learner (Abdelrahman et al., 2023). This assumption becomes less straightforward when learning takes place through collaboration.

When learners work together, the task may produce one shared solution rather than separate responses from each learner. Consider two students debugging a program together. One may identify a faulty loop while the other edits the code. The corrected program shows that the pair resolved the problem, but it does not establish what each learner understood. Pair-programming research provides a clear example of work in which responsibility for producing a shared solution can be distributed between partners (Plonka et al., 2011). A similar issue occurs in collaborative writing, where learners may contribute different language knowledge to the same text (Storch, 2005), one suggesting vocabulary and another correcting grammar for example. In both cases, the final product is shared while the knowledge behind it may differ between learners (Von Davier & Halpin, 2013; Webb, 1993; Wilson et al., 2017).

This creates a specific problem for knowledge tracing. Prior work can incorporate information from peers or model individually identifiable behaviour during collaboration (Long et al., 2022; Olsen et al., 2017; Walker et al., 2014), but the observation being modelled can still be associated with a particular learner. Less clear is how one shared correct or incorrect observation should update separate knowledge estimates when learners contributed differently to producing it.

This distinction matters for adaptive support. A system responding only to group success may overlook a learner who relied on a partner, while a system responding to group failure may provide unnecessary support to someone who already understands the relevant skill. Supporting individuals within a group therefore requires a way to maintain separate knowledge estimates without requiring each learner to produce a separate answer (Evans et al., 2019).

We introduce Attribution-based Bayesian Knowledge Tracing (A-BKT) to address this problem. A-BKT retains the shared

correct or incorrect task observation but adjusts how strongly that observation influences each learner's mastery estimate according to their estimated contribution to the evolving solution. Attribution therefore changes the strength of the evidence assigned to a learner, not whether the evidence supports or opposes mastery. A learner who contributes more does not automatically receive a higher mastery estimate. Greater attribution strengthens positive evidence from a correct observation and negative evidence from an incorrect one.

Collaborative tasks also develop over time. Intermediate states may provide useful evidence while learners are constructing a solution, but they may also reflect temporary revisions that should not have the same role as the group's committed answer (Salomons & Scassellati, 2024). We therefore examine both learner-specific attribution and how evidence is handled across the task. The proposed A-BKT model combines learner-specific attribution with running aggregation during the task and a separate task-end commitment. Three comparison models isolate the effects of these two modelling choices.

We evaluate the models offline using data from 13 adult dyads completing collaborative Snap Circuits tasks. Each dyad completed 10 tasks involving 13 main skills. Final mastery estimates were evaluated against individual post-test performance, while estimates during the activity were compared with video annotations of learner-specific evidence. We also examined whether the models distinguished partners who received the same shared observations but had different post-test outcomes for the same skill.

The paper contributes a knowledge-tracing approach for estimating individual knowledge from shared collaborative observations, together with an evaluation of learner-specific attribution and task-level evidence handling. The results provide stronger support for distinguishing between partners and for correspondence with learner-specific evidence during collaboration than for distinguishing higher from lower post-test performance across all learners and skills.

We address two research questions:

- **RQ1:** Does learner-specific attribution improve individual knowledge estimates from shared collaborative observations compared with assigning the evidence equally to all learners?
- **RQ2:** How does separating within-task evidence from the evidence retained at task completion affect individual knowledge estimates, and does the effect of learner-specific attribution depend on how task evidence is handled?

2. Background

Knowledge Tracing Knowledge tracing maintains estimates of a learner's skill mastery across repeated learning opportunities (Abdelrahman et al., 2023). One widely used approach is Bayesian Knowledge Tracing (BKT), which represents each skill as a latent state that is either mastered or unmastered (Corbett & Anderson, 1994).

Let $K \in \{0, 1\}$ denote whether a skill is mastered, where $K = 1$ indicates mastery and $K = 0$ indicates non-mastery. Let $p = P(K = 1)$ denote the current probability of mastery. Let $Y \in \{0, 1\}$ denote the observed skill outcome, where $Y = 1$ indicates a correct observation and $Y = 0$ an incorrect observation. BKT includes a *slip* probability s , representing an incorrect response despite mastery, and a *guess* probability g , representing a correct response despite non-mastery,

$$s = P(Y = 0 \mid K = 1), \quad g = P(Y = 1 \mid K = 0). \quad (1)$$

For an observation Y , the posterior mastery probability after incorporating the observation is

$$p^+ = P(K = 1 \mid Y = 1) = \frac{p(1-s)}{p(1-s) + (1-p)g}, \quad p^+ = P(K = 1 \mid Y = 0) = \frac{ps}{ps + (1-p)(1-g)}. \quad (2)$$

The same observation update can be expressed in log-odds form as

$$\text{logit}(p^+) = \text{logit}(p) + \begin{cases} \log\left(\frac{1-s}{g}\right), & Y = 1, \\ \log\left(\frac{s}{1-g}\right), & Y = 0. \end{cases} \quad (3)$$

Thus, a BKT observation can be viewed as adding the evidence supplied by the observation to the learner's current log-odds of mastery.

BKT also includes a learning-transition probability T , representing the chance that an unmastered skill becomes mastered following a learning opportunity. After the observation update, the mastery estimate is carried forward as

$$p' = p^+ + (1 - p^+)T. \quad (4)$$

Here, p' denotes the mastery probability after applying the learning transition. This value becomes the prior mastery estimate for the next learning opportunity.

Knowledge Tracing in Collaborative Settings Previous work has incorporated information from other learners into individual knowledge tracing. Collaborative Knowledge Tracing (CoKT), for example, combines a learner’s response history with information from students who have similar question-answering histories (Long et al., 2022). Related approaches have also used information from similar learners or learner groups to improve individual knowledge estimates (Liu et al., 2025; Zhang et al., 2024). In these approaches, however, the response being modelled still belongs to an individual learner; information from other learners provides additional context.

Learner modelling has also been applied within collaborative learning environments. BKT has been used to maintain learner-specific estimates from individually identifiable behaviours during peer tutoring (Walker et al., 2014). Other work has modelled individual learning across repeated collaborative practice while accounting for actions involving a partner (Olsen et al., 2017). These approaches show that individual estimates can be maintained during collaboration, but the observations used to update those estimates can still be associated with particular learners.

A different case arises when the task produces a single shared outcome. Group-level performance may not represent the competence of every member (Von Davier & Halpin, 2013; Webb, 1993; Wilson et al., 2017). Collaborative assessment has addressed this problem by modelling a response as jointly produced from the abilities of multiple group members (Halpin & Bergner, 2018). For knowledge tracing, however, the objective is different: separate mastery estimates must be maintained over repeated opportunities even though the observed correct or incorrect state belongs to the group.

This leaves a specific gap. When two learners jointly produce one observation, assigning the complete observation equally to both assumes that it provides the same evidence about their knowledge. Existing collaborative knowledge-tracing work does not directly resolve how the same group-level observation should contribute different amounts of evidence to separate individual mastery traces.

Process Evidence and Learner Contribution Collaborative process data can provide information that is not present in the shared outcome alone. Actions, communication, interaction sequences, and changes to shared artifacts have been used to construct indicators of individual behaviour within collaborative tasks (Care et al., 2016; Harding et al., 2017). Learners within the same group may also take different roles and make different contributions as the task progresses (Volet et al., 2017).

Prior work on collaborative problem solving has analysed individual indicators alongside measures of group performance (Yuan et al., 2019). Collaborative assessment has also incorporated individual contribution when interpreting a shared outcome (Song et al., 2024), while both individual and collective behaviours have been associated with successful collaborative outcomes (Sun et al., 2022). Together, this work shows that the shared outcome and the process that produced it can provide complementary information.

The present work uses this distinction for knowledge tracing. Correctness remains a group-level observation, while learner-specific process information is used to estimate how strongly that observation should be attributed to each learner. The following section describes how this attribution is incorporated into the BKT observation update.

3. Attribution-based Bayesian Knowledge Tracing

A-BKT maintains separate mastery estimates for each learner and skill while using observations produced by the group. The following sections describe how these observations are constructed, attributed to individual learners, and handled across a task.

Observation Construction from Collaborative Process We focus on collaborative tasks in which the evolving solution can be observed, rather than tasks that provide only a final group response. The task process is represented as a sequence of discrete knowledge-tracing observations, referred to as *task states*. In the present study, these states are created at pauses in observable activity, which provide natural boundaries between periods of construction.

For each skill, a task state is first classified as assessable or not assessable. Assessability asks whether the current configuration provides enough information to judge that skill. If it does, the state is classified as correct or incorrect and can update the mastery estimate. If it does not, the state is not used as evidence for that skill. This distinction prevents an incomplete or transitional configuration from being treated automatically as an incorrect demonstration of the skill. The resulting correct and incorrect states form the observation sequence used by the knowledge-tracing model.

Collaborative Slip and Guess In conventional BKT, slip and guess describe the relationship between an individual learner’s mastery and that learner’s observed response. In the present setting, the observation is instead produced by the group. Slip and guess therefore require a collaborative interpretation.

Let $Y \in \{0, 1\}$ denote the shared group observation, where $Y = 1$ indicates a correct state and $Y = 0$ an incorrect state. Let $K_i \in \{0, 1\}$ denote learner i ’s latent mastery state for the relevant skill, where $K_i = 1$ indicates mastery and $K_i = 0$ non-mastery. We define the collaborative slip and guess probabilities as

$$s_c = P(Y = 0 \mid K_i = 1), \quad g_c = P(Y = 1 \mid K_i = 0). \quad (5)$$

An incorrect shared observation can occur even when learner i possesses the relevant skill. For example, the learner's proposed solution may not be implemented, another learner may introduce an error, or the group may make an execution mistake. The collaborative slip parameter s_c therefore captures the possibility that the shared group outcome does not accurately reflect that learner's individual knowledge.

Conversely, a correct shared observation can occur even when learner i does not possess the relevant skill. A partner may provide the correct solution, the learner may follow another member's instruction, or the group may reach the correct state by chance without understanding the relevant skill. The collaborative guess parameter g_c therefore captures the possibility that a correct group outcome does not necessarily provide strong evidence that the learner has mastered the skill.

These parameters capture uncertainty between individual knowledge and the shared outcome. They should not be interpreted as a complete probabilistic model of how the knowledge and actions of all group members jointly produces the observed state. Instead, they provide the base observation evidence that is subsequently weighted by learner-specific attribution.

Let $Y \in \{0, 1\}$ denote the shared group observation, where $Y = 1$ indicates a correct state and $Y = 0$ an incorrect state. Let $K_i \in \{0, 1\}$ denote the latent mastery state of learner i for the relevant skill, $K = 1$ indicating mastery and $K = 0$ indicating non-mastery. We define the likelihood of that observation under the mastered and unmastered states as L_1 and L_0 , respectively:

$$L_1 = P(Y | K_i = 1) = \begin{cases} 1 - s_c, & Y = 1, \\ s_c, & Y = 0. \end{cases} \quad L_0 = P(Y | K_i = 0) = \begin{cases} g_c, & Y = 1, \\ 1 - g_c, & Y = 0. \end{cases} \quad (6)$$

Learner-Specific Attribution We use learner-specific attribution to determine how strongly each shared observation contributes to each learner's mastery estimate.

Let $i \in \{1, \dots, n\}$ index learners in a group of n learners, and let k index shared observations. We denote the attribution of observation k to learner i by $a_{ik} \in [0, 1]$. Attribution is normalized across the group such that

$$\sum_{i=1}^n a_{ik} = 1, \quad \text{and for the dyads in this study} \quad a_{1k} + a_{2k} = 1. \quad (7)$$

A larger value of a_{ik} indicates that a greater proportion of the observable activity leading to observation k is attributed to learner i . Conversely, an attribution close to zero greatly reduces the influence of the observation on that learner's mastery.

Attribution should not be interpreted as a complete measure of cognitive contribution. It describes a particular learner's involvement as captured by observable process signals. The proposed knowledge tracing model *does not* prescribe how a_{ik} must be estimated.

Estimating Attribution in Different Domains A-BKT requires learner-specific attribution values, but their calculation depends on the task and what a contribution means in each context. It could be process information relating to manipulation of task artifacts, verbal interaction, gestures and other non-verbal communication.

We can think of possible ways to supply attribution depending on the domain, even though such methods not are not evaluated in the present study. In collaborative programming, a contribution score could be derived from edits to code relevant to the skill being evaluated. In discussion-based problem solving, similar scores could be derived from proposals, explanations, and revisions that contribute to the group's developing answer. The scoring rules in each case should describe contribution to the observed state, whether that state is correct or incorrect, rather than simply count keystrokes or speaking time.

Importantly, for knowledge tracing in such collaborative settings, the activity should provide repeated opportunities to apply the same identifiable skills. The tasks need not be identical, but their observations must relate to consistent skill definitions so that evidence from successive tasks can inform the same individual knowledge trace.

Attribution-Weighted Observation Evidence As shown in Equation 3, a BKT observation can be interpreted as contributing evidence to the learner's current log-odds of mastery. Using the collaborative likelihoods defined in Equation 6, the evidence supplied by a shared observation is its log-likelihood ratio,

$$\ell = \log \frac{L_1}{L_0} = \begin{cases} \log \left(\frac{1 - s_c}{g_c} \right), & Y = 1, \\ \log \left(\frac{s_c}{1 - g_c} \right), & Y = 0. \end{cases} \quad (8)$$

A-BKT incorporates learner-specific attribution by scaling this evidence. Let $a \in [0, 1]$ denote the attribution assigned to the observation for a particular learner. The attributed observation evidence is

$$e = a\ell. \quad (9)$$

The resulting mastery update is

$$p^+ = \sigma[\text{logit}(p) + e], \quad \text{where} \quad \sigma(z) = \frac{1}{1 + \exp(-z)}. \quad (10)$$

Equivalently, using the likelihoods in Equation 6, the update can be written in probability space as

$$p^+ = \frac{pL_1^a}{pL_1^a + (1-p)L_0^a}. \quad (11)$$

When $a = 1$, the observation contributes the full evidence used by the underlying BKT observation update. Values $0 < a < 1$ reduce the magnitude of the update. When $a = 0$, the observation contributes no evidence and the mastery estimate remains unchanged.

Evidence Across the Task Process The observations produced during a task do not all play the same role. Intermediate states can provide useful evidence while a solution is being developed, but they may also reflect temporary revisions (Salomons & Scassellati, 2024). So the evidence the intermediate states provide can be transient or noisy, than a response that has been explicitly committed as the final answer.

Within-task estimation is useful because individual support may be needed while learners are still working, rather than only after a final answer has been submitted. At the same time, knowledge tracing must carry an estimate forward when the same skill is encountered again in later tasks. The model must therefore support two needs. First, it must keep an updated estimate while the current task is still progressing. Second, when the task ends, it must determine the estimate to be carried forward as the prior for the next task. A temporary intermediate configuration should not necessarily have the same role in this carry-over as the group's committed final answer.

We therefore distinguish between two estimates: a *provisional within-task estimate*, used to track mastery during a task, and a *committed task-end estimate*, that is retained after task completion. Provisional estimates track how the model changes as evidence appears during the task. The committed estimate becomes the starting point for the next relevant task.

Running Aggregation During a task, each observation contributes evidence to a provisional mastery estimate. As new observations become available, the model recalculates this estimate using the evidence observed so far. This provides a continuous estimate of mastery while the shared solution is still being developed.

Intermediate states are treated as repeated views of an evolving solution. Their mean attributed evidence is therefore used for the provisional estimate, rather than allowing the strength of the evidence to increase simply because more intermediate states were recorded. This prevents the magnitude of the provisional evidence from increasing solely because a task contains more recorded intermediate states. Repeating the same evidence leaves its mean unchanged, while changes in the balance of correct and incorrect evidence can still change the estimate.

Let $e_{r,j}$ denote the evidence attributed observation j in task r for a certain learner. After k usable observations, the mean attributed evidence is

$$\bar{e}_{r,k} = \frac{1}{k} \sum_{j=1}^k e_{r,j}. \quad (12)$$

Let p_r^{entry} denote the learner's mastery estimate at the beginning of task r . The provisional estimate after k observations is

$$p_{r,k}^{\text{prov}} = \sigma(\text{logit}(p_r^{\text{entry}}) + \bar{e}_{r,k}), \quad (13)$$

Each provisional estimate is recalculated from the task-entry mastery estimate. The mean evidence is not recursively added to the previous provisional estimate. The provisional estimate should therefore be treated as a working estimate of mastery while the task is still in progress.

At task completion, the provisional estimate is replaced by a separate committed estimate calculated from the task-entry mastery estimate. This update uses the task-end shared observation together with a task-level attribution value summarizing the learner's contribution across the task.

Let $\bar{a}_r$ denote the task-level attribution of a learner in task r and where E_r denotes the set of events for which attribution is available. When event-level attribution is available, this value is the mean of the attribution values observed during the task:

$$\bar{a}_r = \frac{1}{|E_r|} \sum_{j \in E_r} a_{r,j} \quad (14)$$

Let $Y_{r,end} \in \{0, 1\}$ and $\ell_{Y_{r,end}}$ denote the observation and its log-likelihood at the end of task r . The committed estimate is

$$p^{\text{commit}} = \sigma \left(\text{logit} (p^{\text{entry}}) + \bar{a}_r \ell_{Y_{r,end}} \right). \quad (15)$$

The task-end state summarizes the group outcome, while $\bar{a}_r$ summarizes the learner's attribution across the task. Because the committed update begins from the task-entry estimate, intermediate states can inform within-task tracking without being accumulated into the value carried forward. The standard BKT learning transition is then applied once, and the resulting mastery estimate becomes the entry estimate for the next relevant task.

In settings where event-level process information is unavailable, the same task-end update can still be used with a task-level attribution value and task-end outcome. Only continuous within-task estimation requires richer event-level observations and their attribution.

A-BKT and Comparison Models The proposed A-BKT model combines learner-specific attribution with running aggregation. To examine the contribution of each modelling choice, we compare A-BKT with three comparison models. Together, the four models form a 2×2 design: attribution is either equal or learner-specific, and evidence handling is either reactive or based on running aggregation. All four use the same collaborative observation evidence described above and apply a single learning transition at task completion.

Reactive Equal Attribution (R-EA) R-EA provides the baseline comparison. It assigns the same proportion of each shared observation to every learner. For a group of n learners, the attribution weight is $1/n$. In the present study, participants worked in dyads, so each learner receives a weight of $1/2$, or 50%.

R-EA handles observations reactively. Each usable observation is incorporated sequentially, so evidence from earlier observations remains in the mastery estimate as new observations are received. Using the attributed evidence defined in Equation 9, the mastery estimate after k usable observations in task r is

$$p_{r,k}^{\text{prov,reactive}} = \sigma \left(\text{logit} (p_r^{\text{entry}}) + \sum_{j=1}^k e_{rj} \right), \quad (16)$$

where equal attribution gives $e_{rj} = \frac{1}{n} \ell(Y_{rj})$. At task completion, the final accumulated estimate is retained. No separate task-end evidence recalculation is performed.

Reactive Learner-Specific Attribution (R-A) R-A uses the same reactive evidence handling as R-EA, but replaces equal attribution with the observed learner-specific attribution a_{rj} . The evidence contributed by an observation is therefore $e_{rj} = a_{rj} \ell(Y_{rj})$. A shared observation can consequently influence the two learners' mastery estimates by different amounts. Evidence still accumulates sequentially throughout the task, and the final accumulated estimate is retained at task completion. Comparing R-A with R-EA isolates the effect of learner-specific attribution under reactive updating.

Running-Aggregation Equal Attribution (RA-EA) RA-EA retains equal attribution but changes how evidence is handled across the task. During task completion, it uses the mean attributed evidence defined in Equation 13 to maintain a provisional mastery estimate. Each provisional estimate is recalculated from the task-entry mastery estimate rather than from the previous provisional estimate. At task completion, the provisional estimate is not carried forward. Instead, a separate committed estimate is calculated from the task-entry mastery estimate using the task-end shared observation. Because attribution remains equal, the task-end observation is weighted by $1/n$ for each learner. Comparing RA-EA with R-EA isolates the effect of running aggregation and task-end commitment while keeping attribution equal.

Running-Aggregation Learner-Specific Attribution (RA-A; A-BKT) RA-A combines learner-specific attribution with running aggregation and task-end commitment and is the proposed *Attribution-based Bayesian Knowledge Tracing* (A-BKT) model. During the task, A-BKT uses the learner-specific attributed evidence $e_{rj} = a_{rj} \ell(Y_{rj})$ within the running mean to maintain a provisional mastery estimate. At task completion, this provisional estimate is not carried forward. A separate committed estimate is instead calculated from the task-entry mastery estimate using the task-end shared observation, weighted by the learner's task-level attribution $\bar{a}_r$.

A-BKT therefore allows the same shared observation to provide different amounts of evidence about each learner, while running aggregation separates the evidence used to track mastery during the task from the evidence retained at task completion.

Comparing RA-A with RA-EA isolates the effect of learner-specific attribution under running aggregation, while comparing A-BKT with R-A isolates the effect of the evidence-handling approach under learner-specific attribution.

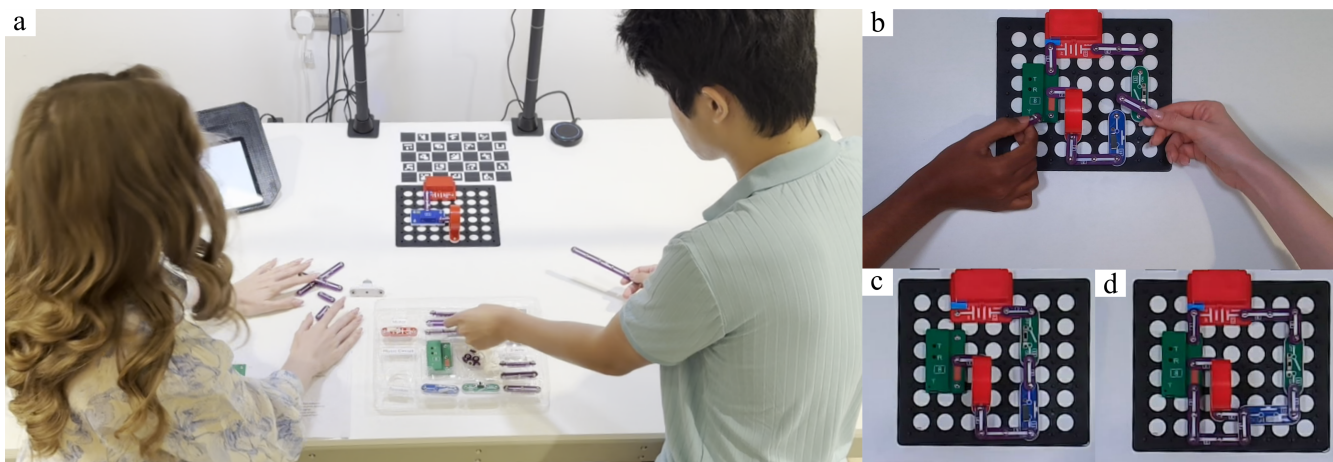


Figure 1. Collaborative Snap Circuits task and observation process. (a) Overall collaborative setup. (b) Participants working on the shared circuit. (c) Intermediate evaluated task state. (d) Completed task state. Participants worked together on a shared physical circuit board. Component manipulations provided learner-specific process information, while unobstructed board configurations were evaluated as shared task states during and at the end of the task.

4. User Study

We conducted a user study with two-person groups to collect collaborative process and outcome information for offline model evaluation. The chosen task, Snap Circuits, provided repeated observations for comparing different ways of handling evidence within a task. All dyads followed the same study procedure. The knowledge tracing models were applied after data collection and did not influence the tasks, feedback, or interaction experienced by participants.

Participants Thirty adults participated in the study, forming 15 dyads. Each dyad produced one shared collaborative interaction stream, from which separate individual knowledge traces were later estimated. Two dyads were excluded from the analysis because occlusions in their video recordings prevented reliable learner-specific annotation. Therefore the analyzed sample comprised 26 participants in 13 dyads.

The sample included 16 females and 14 males, with most participants aged between 21 and 29 years. Participants were recruited through university advertisements and came from a range of disciplinary backgrounds. All had either completed or were currently pursuing at least a bachelor's degree. No prior knowledge of electronic circuits was required. Participants were required to be at least 18 years old and able to communicate in English.

Participants were paired according to overlapping availability. Members of three dyads knew each other well before the study, while the remaining participants had not previously met their partner. No familiarization period was provided before the collaborative activity. Participants were compensated for their time. The study was approved by the institutional ethics committee, and all participants provided informed consent before taking part.

Collaborative Tasks and Skills The collaborative activity used Snap Circuits (Elenco Electronics, 2026). Two participants worked together on one shared board to construct circuits satisfying a stated task goal, as shown in Figure ???. The physical nature of the activity made both the developing shared solution and many of the participants' individual actions observable.

Each dyad completed 10 circuit-building tasks. Different dyads received different combinations of tasks, while the frequency the skills were encountered was kept approximately similar across groups. Skills repeated across tasks so that evidence about the same knowledge could be observed on multiple occasions.

The activity covered 13 main circuit skills. These included knowledge of component roles such as sources, loads, and switches; connecting components through appropriate power and output terminals; using polar and non-polar components; and constructing basic logical relationships including AND, OR, and NOT. Tasks increased gradually in difficulty according to the skills required to complete them. Depending on difficulty, each task was allocated between 3 and 5 minutes.

Study Procedure Each session lasted approximately one hour. Participants first completed informed consent and a demographic questionnaire, followed by an individual pre-test. They then completed the collaborative circuit-building activity, followed by an individual post-test.

Before the collaborative activity, participants were shown how the Snap Circuit pieces physically attach to the board. This demonstration covered only the mechanics of using the materials and did not teach the circuit concepts assessed in the study. Both participants worked on the same circuit board and were instructed to collaborate naturally. No roles were assigned, and

either participant could discuss or modify the developing solution. For each task, the interface displayed the task goal and the remaining time. For example, a task could ask the group to build a circuit that plays radio through a speaker when a press switch is activated. Participants used a submit button to submit the current circuit for evaluation.

While time remained, participants could submit multiple times. After each submission, the interface reported only whether the complete circuit was correct or incorrect. It did not identify which component, connection, or skill was responsible for an incorrect answer. Following an incorrect submission, the group could continue revising the circuit and submit again. A correct submission ended the task immediately. If the time limit expired first, the group was allowed one final submission before moving on. No instructional hints or skill-level feedback were provided. The experimenter was not present in the room during the main collaborative activity.

For the knowledge-tracing model, only the submission that ended the task defined the task-end shared observation used for the committed estimate. Earlier submissions remained part of the within-task process.

Individual Pre- and Post-Test The pre- and post-tests provided individual measures of circuit knowledge before and after the collaborative activity. Both assessments were completed individually using pen and paper, with members of each dyad separated during the assessment.

Each test contained six questions covering all 13 main skills. For each question, participants were shown four circuit configurations and asked to identify all configurations that correctly satisfied the stated requirement. More than one configuration could be correct. For configurations identified as incorrect, participants were also asked to select applicable reasons for the error from a predefined list.

The pre- and post-tests assessed the same skills but did not present identical answer configurations. Components were repositioned, circuit orientations were changed, and alternative valid circuits were given as answer choices. This reduced the likelihood of participants memorizing the pre-test. The assessments used the same visual representation of the board and components as the collaborative activity. No correctness feedback was provided during or after either assessment.

Process Data and Attribution Estimation The collaborative activity recorded using the overhead Kinect camera fed into computer-vision algorithm that tracked the developing circuit and detected changes to components on the shared board. It also tracked the participants' hands so that observed component changes could be associated with the participant who performed them. The resulting process record contained the timing of component additions, removals, and other changes together with their associated participant.

The continuous process record was segmented into task states using pauses in activity as boundaries between periods of construction. For each skill, the circuit at each state was classified as correct, incorrect, or not assessable. A state was not assessable when the current configuration did not provide sufficient evidence to judge the skill as correct or incorrect.

Learner-specific attribution for each shared observation was estimated from the component actions occurring between the preceding and current task states. Actions occurring closer to the current state were given greater weight than actions occurring earlier in the interval, on the assumption that more recent actions are more representative of the configuration being evaluated. The weighted contributions were combined for each learner and normalized across the dyad so that the two attribution values summed to one. Attribution therefore represents each learner's observable contribution to producing the shared state. It does not directly measure the learner's knowledge or determine who proposed or decided the underlying change.

Manual Annotation of Individual Skill Evidence Video recordings of the collaborative sessions were used to obtain an independent individual-level reference during task completion. Selected task states were annotated separately for each learner and skill. At each state, the annotator considered the interaction observed up to that point and asked: *Does sufficient evidence exist that this learner has demonstrated the skill, and does that evidence indicate a correct or incorrect demonstration?*

When sufficient evidence was available, the state was labelled as a correct demonstration (1) or an incorrect demonstration (0). States for which the learner's behaviour did not provide sufficient evidence were left unlabelled. These unlabelled states were not treated as either correct or incorrect observations.

The annotations were described the evidence demonstrated by an individual learner at a particular point in the collaborative process. They were not treated as direct observations of a learner's latent mastery. Annotators considered only information available up to the annotated state and did not use future behaviour or model estimates when assigning labels. To assess annotation reliability, a second annotator independently coded 20% of the annotation units mixing different groups and tasks.

5. Results

The analyses included 26 participants from 13 dyads. Model performance was evaluated against individual post-test outcomes and video annotations of learner-specific evidence during the tasks. We also examined whether the models distinguished partners with different post-test outcomes.

Pre-Post Skill Performance Each of the 13 skills was scored as either demonstrated (1) or not demonstrated (0). For each participant, overall performance was calculated as the proportion of the 13 skills demonstrated. The mean proportion of skills demonstrated increased from 0.436 ($SD = 0.188$) at pre-test to 0.591 ($SD = 0.214$) at post-test, corresponding to a mean increase of 0.155. Of the 26 participants, 21 improved, one showed no change, and four had lower post-test scores. Figure 2 shows the individual pre-post changes.

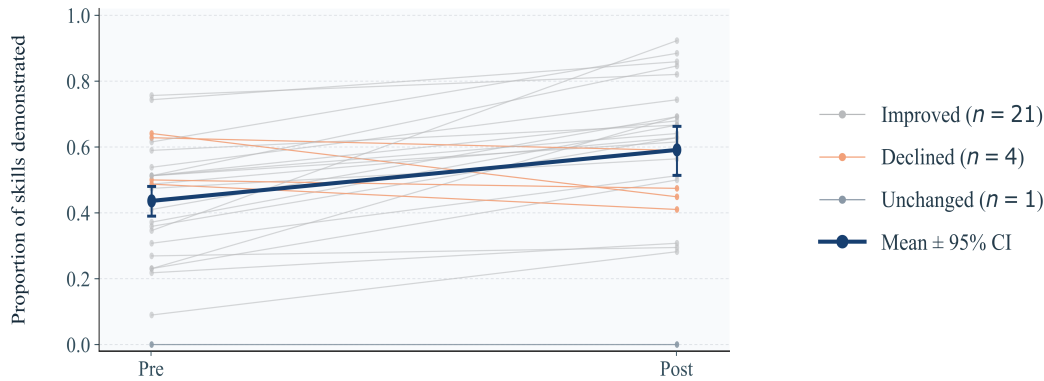


Figure 2. Proportions of skills demonstrated at pre-test and post-test. Thin lines connect each participant's scores; navy points and error bars show the mean and 95% whole-dyad bootstrap confidence intervals.

Model Evaluation A-BKT and the three comparison models were evaluated using leave-one-dyad-out cross-validation. Each of the 13 dyads was reserved for evaluation in turn, while model parameters were fitted using the shared task outcomes from the other 12 dyads. These parameters were then kept fixed while the model processed the reserved dyad's observations to estimate each learner's mastery. Both partners were therefore excluded from parameter fitting when their estimates were evaluated. Individual post-test scores and video annotations were not used to fit the parameters.

For each model, the resulting held-out estimates were combined across the 13 dyads. Final AUROC compared final mastery estimates with binary post-test values, while within-task AUROC compared post-observation estimates with binary annotation labels. Each AUROC was calculated from all eligible cases combined, rather than by averaging separate dyad-level AUROCs.

Illustrative Mastery Progression Figure 3 shows estimates from A-BKT (RA-A) and the three comparison models for one learner and skill across five tasks, alongside the shared observations and learner-specific attribution. All models began with a mastery estimate of 0.5.

During the first two tasks, estimates decreased following incorrect observations and increased when correct states appeared. By the end of Task 2, both equal-attribution models were close to 1, whereas the learner-specific attribution models remained lower. The incorrect observation at the end of Task 3 produced a visible decrease in R-A and RA-A, while the equal-attribution estimates remained close to 1.

Task 4 shows a clearer difference between reactive updating and running aggregation. The shared state changed from incorrect to correct and remained correct, while the learner's attribution fell from near 1 early in the task to near 0 later. R-A increased towards 1 as correct observations accumulated. The provisional RA-A estimate increased more gradually and remained below R-A. Both equal-attribution estimates stayed close to 1 through most of this task.

The task-end markers also show the distinction between provisional and committed estimates. At the end of Task 4, RA-A's committed estimate was higher than its final provisional estimate, followed by a further increase from the learning transition. This post-learning estimate was carried into Task 5. By the end of the sequence, both reactive models and RA-EA were close to 1, while RA-A remained lower.

This example illustrates differences in how the models responded to the same collaborative observations. The following analyses evaluate their correspondence with individual post-test outcomes and learner-specific evidence identified in the video annotations.

Final Individual Mastery Estimates Final evaluation included 338 learner-skill cases across the 13 main skills. Across the 26 learners, this resulted in 258 skills scored as demonstrated and 80 scored as not demonstrated.

Both the final and within-task AUROC analyses combined all eligible held-out mastery estimates and their corresponding binary values. Within each analysis, the same cases were evaluated for all four models. Figure 4 presents model performance with 95% confidence intervals, calculated from 10,000 bootstrap samples of whole dyads. Table 1 reports the paired comparisons. All model-comparison p -values reported below are Holm-adjusted.

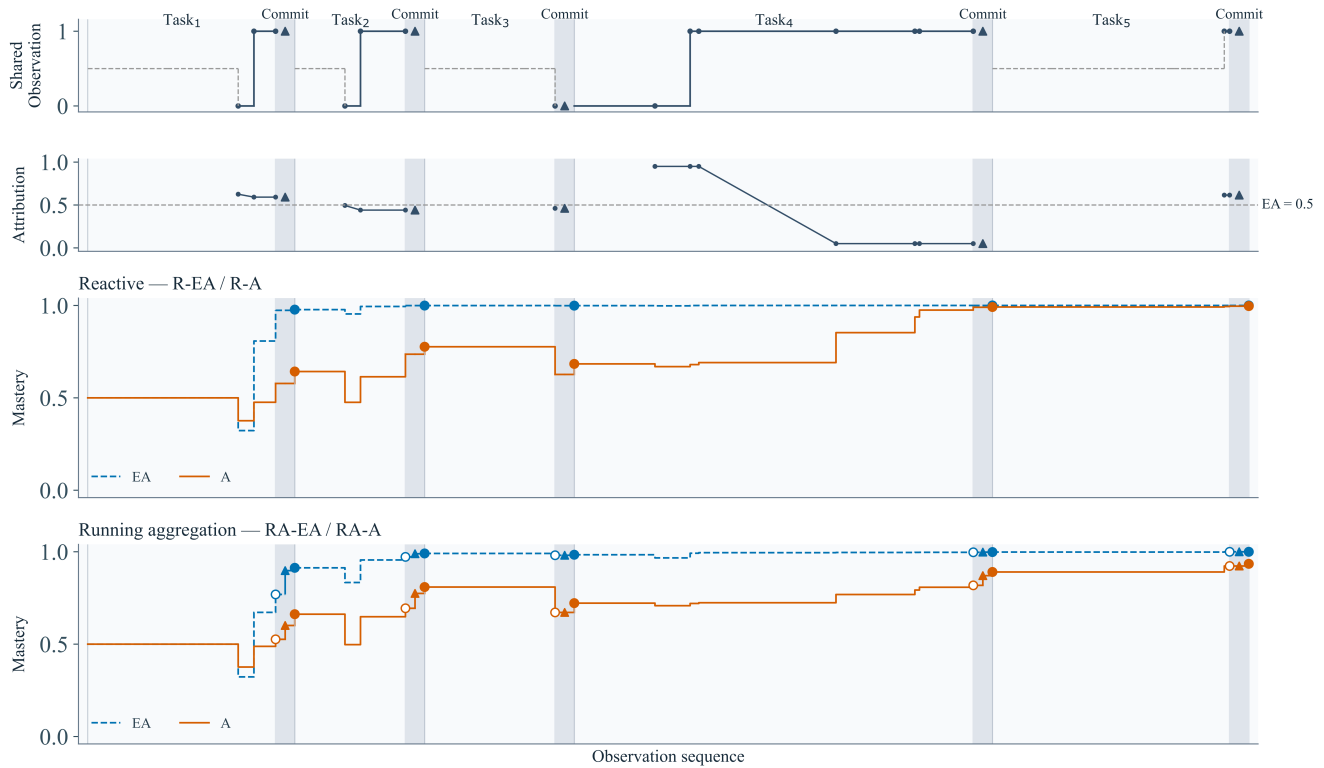


Figure 3. Illustrative mastery progression for one learner and skill across five tasks. Upper panels show group observations and learner attribution; grey dotted segments in the observation panel indicate unassessable states, not a value of 0.5. Blue dashed curves use equal attribution; orange curves use learner-specific attribution. Triangles in the upper panels mark the task-end observation and attribution used at task end. For running aggregation, open circles mark final provisional estimates and triangles mark committed estimates before learning. Filled circles in both mastery panels mark estimates after learning. Shaded task-end regions do not represent elapsed time; post-learning mastery carries into the next task.

Final AUROC was 0.422 (95% CI [0.347, 0.508]) for R-EA and 0.474 ([0.398, 0.557]) for R-A. The corresponding values were 0.465 ([0.399, 0.535]) for RA-EA and 0.527 ([0.457, 0.595]) for RA-A. All four confidence intervals included 0.5.

Learner-specific attribution increased final AUROC by 0.052 under reactive updating (95% CI [0.027, 0.073]) and by 0.062 under running aggregation ([0.029, 0.091]). Neither comparison was statistically significant after correction (both $p_{\text{adj}} = 0.0513$). Running aggregation also produced numerically higher AUROC than reactive updating, by 0.043 with equal attribution and 0.053 with learner attribution, but neither difference was statistically significant (both $p_{\text{adj}} = 0.634$). The interaction between attribution and evidence handling was not significant ($\Delta\text{AUROC} = 0.010$, 95% CI [-0.011, 0.029], $p_{\text{adj}} = 0.535$).

Within-Task Individual Mastery Estimates The within-task evaluation used video annotations indicating whether the observed interaction provided sufficient learner-specific evidence of correct or incorrect skill use. In the second-annotation sample, 184 units received an explicit binary label from both annotators. Agreement was 89.7%, with Cohen's $\kappa = 0.620$, calculated before disagreements were resolved. Units without an explicit label from either annotator were excluded from this agreement calculation.

Of the 1,430 binary annotations, 1,372 could be matched to model estimates across all four models. The remaining 58 were excluded because a unique matching event was unavailable. The evaluated sample covered all 26 participants, comprising 1,014 correct and 358 incorrect annotation labels. These labels were compared with post-observation mastery estimates before the task-end learning transition.

Post-update annotation AUROC was 0.708 (95% CI [0.670, 0.744]) for R-EA and 0.721 ([0.680, 0.763]) for R-A. Running aggregation produced values of 0.811 ([0.779, 0.838]) for RA-EA and 0.829 ([0.796, 0.860]) for RA-A (Figure 4).

Under reactive updating, learner attribution increased AUROC by 0.013 (95% CI [0.003, 0.028]), but the difference was not statistically significant after correction ($p_{\text{adj}} = 0.129$). Under running aggregation, learner attribution increased AUROC by 0.018 (95% CI [0.008, 0.031], $p_{\text{adj}} = 0.031$).

Running aggregation had higher annotation AUROC than reactive updating under both attribution conditions. The increase was 0.103 with equal attribution (95% CI [0.082, 0.121], $p_{\text{adj}} = 0.0012$) and 0.108 with learner attribution (95% CI

Table 1. Pairwise comparisons among the four models. Each cell reports the first model minus the second, Δ , its pointwise 95% whole-dyad bootstrap confidence interval, and Holm-adjusted p -value. AUROC differences are in AUROC units; individual differentiation differences are in percentage points. For each AUROC, Holm correction covers the four contrasts and the interaction; for individual differentiation, it covers the four contrasts.

Metric	Attribution effect		Evidence-handling effect	
	R-A vs R-EA	RA-A vs RA-EA	RA-EA vs R-EA	RA-A vs R-A
Final AUROC	$\Delta = +0.052$ [0.027, 0.073] $p_{\text{adj}} = 0.0513$	$\Delta = +0.062$ [0.029, 0.091] $p_{\text{adj}} = 0.0513$	$\Delta = +0.043$ [0.018, 0.064] $p_{\text{adj}} = 0.6343$	$\Delta = +0.053$ [0.023, 0.076] $p_{\text{adj}} = 0.6343$
Annotation AUROC	$\Delta = +0.013$ [0.003, 0.028] $p_{\text{adj}} = 0.1294$	$\Delta = +0.018$ [0.008, 0.031] $p_{\text{adj}} = 0.0308$	$\Delta = +0.103$ [0.082, 0.121] $p_{\text{adj}} = 0.0012$	$\Delta = +0.108$ [0.085, 0.128] $p_{\text{adj}} = 0.0012$
Individual differentiation	$\Delta = +29.3$ [19.2, 38.4] $p_{\text{adj}} = 0.0059$	$\Delta = +34.5$ [24.5, 44.2] $p_{\text{adj}} = 0.0039$	$\Delta = 0.0$ [0.0, 0.0] $p_{\text{adj}} = 1.0000$	$\Delta = +5.2$ [-5.4, 15.8] $p_{\text{adj}} = 1.0000$

[0.085, 0.128], $p_{\text{adj}} = 0.0012$). The interaction was not statistically significant ($\Delta\text{AUROC} = 0.005$, 95% CI [-0.004, 0.013], $p_{\text{adj}} = 0.385$).

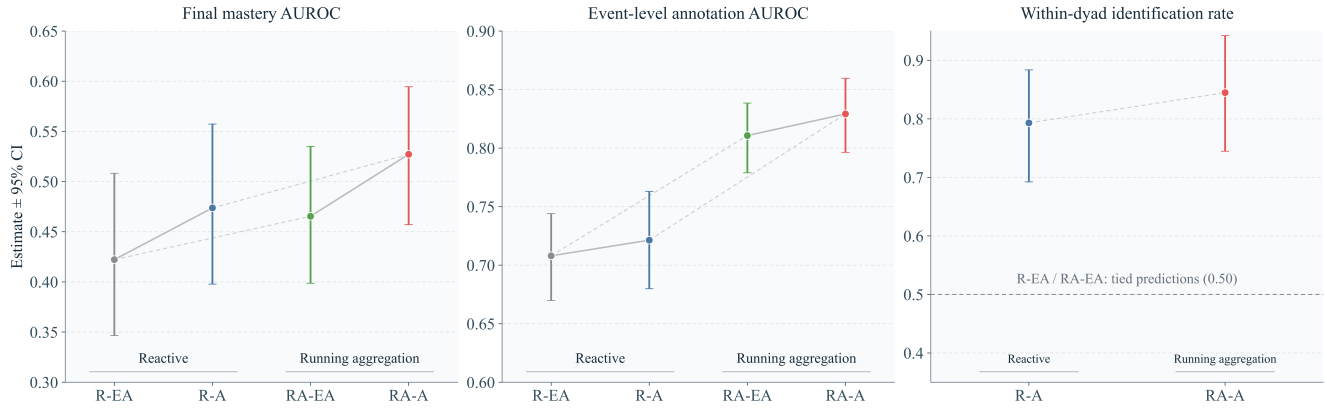


Figure 4. Held-out model performance. Points show metric estimates; whiskers show 95% whole-dyad bootstrap confidence intervals. Identification is evaluated only on dyad-skill pairs with different post-test outcomes. Its panel shows learner-specific attribution models and the 0.50 reference for both equal-attribution models, which assigned tied estimates and received half credit. Solid connectors link attribution conditions; dashed connectors link evidence-handling strategies. Vertical scales differ across panels.

Within-Dyad Identification We also examined whether the models distinguished the two partners when their post-test outcomes differed for the same skill. Of the 169 dyad-skill pairs, 111 (65.7%) had matching outcomes and 58 (34.3%) had different outcomes. The latter came from all 13 dyads and formed the identification sample. Matching-outcome pairs remained included in the final AUROC analysis.

A pair received a score of 1 when the learner with a post-test outcome 1 had the higher final mastery estimate, 0.5 when the estimates were tied, and 0 when the ordering was reversed. The identification score was

$$I = 100 \frac{N_{\text{correct}} + 0.5N_{\text{tied}}}{N_{\text{pairs}}}, \quad (17)$$

where N_{correct} and N_{tied} are the numbers of correctly ordered and tied pairs, respectively, and $N_{\text{pairs}} = 58$.

As expected with identical initial mastery and equal evidence allocation, both equal-attribution models assigned identical estimates to the partners in all 58 pairs. Both therefore scored 50.0%, receiving half credit for every tied pair. R-A correctly ordered 46 pairs and reversed 12, giving a score of 79.3% (95% CI [69.2, 88.4]). RA-A correctly ordered 49 pairs and reversed nine, giving a score of 84.5% (95% CI [74.5, 94.2]). Neither learner-specific attribution model produced tied estimates.

Compared with equal attribution, learner attribution increased identification by 29.3 percentage points under reactive updating (95% CI [19.2, 38.4], $p_{\text{adj}} = 0.006$) and by 34.5 percentage points under running aggregation (95% CI [24.5, 44.2], $p_{\text{adj}} = 0.004$). RA-A exceeded R-A by 5.2 percentage points, but this difference was not statistically significant (95% CI [-5.4, 15.8], $p_{\text{adj}} = 1.000$; Table 1).

6. Discussion

The two modelling choices contributed in different ways. Learner-specific attribution provided the clearest benefit for distinguishing partners, while running aggregation showed its clearest advantage in correspondence with learner-specific evidence during the task. Evidence for improved final post-test discrimination was more limited.

Effect of Learner-Specific Attribution (RQ1) R-A and A-BKT (RA-A) correctly ordered the two learners in 79.3% and 84.5% of these pairs, respectively, assigning the higher estimate to the learner with the positive post-test outcome. Equal attribution produced tied estimates because both partners started with the same mastery estimate and received the same evidence. They provide a non-differentiating reference for this analysis, rather than a competing method capable of identifying individual differences under these settings. The important finding is therefore not merely that attribution separated the estimates, but that their ordering usually agreed with the individual assessment.

Attribution had smaller effects on the two AUROC measures. It improved annotation AUROC under running aggregation, while the increase under reactive updating was not statistically significant after correction. Final AUROC also increased under both strategies, but neither comparison met the adjusted significance threshold. The results therefore support attribution most directly as a means of distinguishing partners, rather than establishing an improvement across every measure of individual knowledge estimation.

These results concern the allocation of evidence, not a direct equation between participation and knowledge. Greater attribution strengthens positive evidence from a correct observation and negative evidence from an incorrect observation. A learner is not assigned greater mastery simply because they perform more actions.

Effect of Running Aggregation (RQ2) Running aggregation improved annotation AUROC under both attribution conditions. This supports the combined strategy of maintaining a provisional estimate during construction and making a separate update at task completion. Its advantage was clearest for agreement with the within-task annotations. The comparisons did not establish an improvement in final AUROC or in identification between A-BKT (RA-A) and R-A. The interaction was also not statistically significant, so the results did not establish that the effect of learner-specific attribution depended on how task evidence was handled.

The within-task finding is consistent with the difference between a developing solution and a committed answer. Intermediate states may reflect revision rather than separate attempts to demonstrate knowledge. Repeatedly accumulating them can give temporary configurations a lasting influence on the estimate, a concern also identified in previous work (Salomons & Scassellati, 2024). The provisional and committed distinction provides a way to use this evidence during the task without retaining every intermediate update afterwards.

Using mean evidence prevents its magnitude from growing directly with the number of repeated states. It does not make the estimate independent of the recording process, since repeating particular states can still change the mean. Moreover, running aggregation changes both within-task estimation and task-end carry-over. The comparison therefore supports the combined strategy, not averaging alone. It also does not establish that a submitted group answer is reliable evidence of every member's knowledge.

Within-Task and Final Performance The strong within-dyad identification and weak final AUROC are not contradictory. Identification compares two partners on the same skill, whereas final AUROC compares positive and negative cases across all learner-skill cases. A model may correctly order two partners while placing both estimates above or below those of other learners inappropriately. Identifying the higher-scoring partner therefore does not establish reliable ranking across groups and skills.

About two-thirds of the dyad-skill pairs had matching post-test outcomes. These pairs provide no observed ordering for the identification analysis, but remain relevant to the overall evaluation. Matching outcomes also do not establish identical knowledge. The identification findings apply to the subset with different outcomes and should not be used to dismiss the weak final AUROC results.

The annotation analysis has a different interpretation again. It compares the estimates with learner-specific evidence coded from the same activity that supplies the model observations. Because each estimate was evaluated after the corresponding event, the analysis measures agreement with contemporaneous evidence about the learner's skill use, not prediction of a future individual response. Annotations were not used for parameter fitting, but they are not direct observations of latent mastery. Unlabelled moments were also excluded, so the analysis does not establish how well the model handles cases where individual evidence is insufficient. All four final AUROC confidence intervals included 0.5, so the results did not establish discrimination above chance across all learner-skill cases.

Limitations Physical manipulation was used to estimate attribution because learners could show their partner a proposed change by making it on the shared board. This made manipulation a useful record of observable contribution, although the person performing an action was not necessarily the person who proposed it. In some dyads, one learner instructed the other to

select or place a component. These informally observed episodes distinguish *action execution* from *decision contribution*, but their frequency was not systematically measured.

Unequal participation should likewise not be interpreted automatically as dominance or as a stable personality trait. A learner may take the lead because they understand the task, guide their partner, or limit their partner's opportunity to act. Previous research distinguishes expertise, participation, and influence, and shows that influence may reflect confidence as well as accuracy (Bonner & Bolinger, 2013; Mayo et al., 2020). The present attribution signal cannot distinguish these explanations. Nor can it fully capture a knowledgeable learner who observes without manipulating the circuit or a learner who acts by following instructions.

Physical access may also affect attribution. A participant closer to the component tray or a particular part of the board may handle more pieces because they are easier to reach. Research on tabletop work shows that spatial organization helps structure collaboration and resource use (Scott et al., 2004). We did not quantify this possible influence. Varying seating and component positions would help test whether attribution reflects contribution beyond differences in access to materials.

Only dyads were evaluated. Normalizing attribution across more learners does not establish that the same signal will remain informative in larger groups. A randomized practical-skills study found less individual hands-on time and more passive activity as group size increased, without significantly poorer transfer performance (Noerholk et al., 2022). This suggests that manipulation-based attribution may be easier to interpret in smaller groups, where fewer contributors must be distinguished. This is not a claim that smaller groups learn better or that the model has an established maximum group size.

The analysis included only 13 independent dyads in one task domain. Excluding two dyads because of video occlusion also restricted evaluation to interactions that could be annotated reliably. Many recorded events do not substitute for more independent groups. The observed pre-post increase cannot be attributed to the models, which were applied offline and did not affect participants' experience. Without a control condition, it also cannot be attributed specifically to collaboration rather than practice, feedback, or assessment familiarity.

Future Work A first step is to measure when physical attribution misses other forms of contribution. A behavioural coding study could record proposals, verbal instructions, pointing, handovers, and corrections alongside the person executing each action. This would establish how frequently action execution and decision contribution differ, and when board interaction alone provides sufficient attribution information. Participation frameworks that distinguish cognitive, coordinative, and other forms of involvement could guide this analysis (Feng & Wei, 2025).

The next step would be to combine manipulation with speech and gesture, adding other signals where they address a specific ambiguity. Multimodal learning analytics provides a basis for investigating these combinations (Blikstein & Worsley, 2016). Comparisons with manipulation alone should establish whether the additional signals improve individual estimates. Uncertain attribution should remain uncertain rather than forcing an action or decision to belong to one learner.

A study that involves a robot tutor could use the estimates to guide learners in similar collaborative tasks. The robot might ask a particular learner to explain a connection or offer a targeted hint, rather than providing the same support to the whole group. Comparing learner-specific and group-level support would test whether the estimates improve individual learning while preserving collaboration. Independent assessments, delayed tests, and observations of participation would evaluate both learning and interaction effects. This would test the practical value of acting on the estimates, which the present offline study does not establish.

References

- Abdelrahman, G., Wang, Q., & Nunes, B. (2023). Knowledge tracing: A survey. *ACM Computing Surveys*, 55(11), 1–37. <https://doi.org/10.1145/3569576>
- Blikstein, P., & Worsley, M. (2016). Multimodal learning analytics and education data mining: Using computational technologies to measure complex learning tasks. *Journal of learning analytics*, 3(2), 220–238.
- Bonner, B. L., & Bolinger, A. R. (2013). Separating the confident from the correct: Leveraging member knowledge in groups to improve decision making and performance. *Organizational Behavior and Human Decision Processes*, 122(2), 214–221. <https://doi.org/10.1016/j.obhdp.2013.07.005>
- Care, E., Scoular, C., & Griffin, P. (2016). Assessment of collaborative problem solving in education environments. *Applied Measurement in Education*, 29(4), 250–264. <https://doi.org/10.1080/08957347.2016.1209204>
- Corbett, A. T., & Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction*, 4(4), 253–278. <https://doi.org/10.1007/BF01099821>
- Elenco Electronics. (2026). *Snap Circuits*. <https://elenco.com/snapsystems/>
- Evans, A., Davis, K., & Wobbrock, J. (2019). Adaptive support for collaboration on tabletop computers.

- Feng, S., & Wei, Z. (2025). The compas framework for modeling individual participation in collaborative learning processes: A systematic review. *International Journal of Computer-Supported Collaborative Learning*, 20(4), 425–461. <https://doi.org/10.1007/s11412-025-09450-x>
- Halpin, P. F., & Bergner, Y. (2018). Psychometric models of small group collaborations. *Psychometrika*, 83(4), 941–962. <https://doi.org/10.1007/s11336-018-9631-z>
- Harding, S.-M. E., Griffin, P. E., Awwal, N., Alom, B. M., & Scoular, C. (2017). Measuring collaborative problem solving using mathematics-based tasks. *AERA Open*, 3(3), 2332858417728046. <https://doi.org/10.1177/2332858417728046>
- Liu, X., Hu, P., & Su, Y. P. (2025). Integrating individualised and similar group in knowledge tracing. *Computers and Electrical Engineering*, 123, 110105. <https://doi.org/10.1016/j.compeleceng.2025.110105>
- Long, T., Qin, J., Shen, J., Zhang, W., Xia, W., Tang, R., He, X., & Yu, Y. (2022). Improving knowledge tracing with collaborative information. *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*, 599–607. <https://doi.org/10.1145/3488560.3498374>
- Mayo, A. T., Woolley, A. W., & Chow, R. M. (2020). Unpacking participation and influence: Diversity’s countervailing effects on expertise use in groups. *Academy of Management Discoveries*, 6(2), 300–319. <https://doi.org/10.5465/amd.2018.0044>
- Noerholk, L. M., Morcke, A. M., Kulasegaram, K., Nørgaard, L. N., Harmsen, L., Andreassen, L. A., Pedersen, N. G., Johnsson, V., Vamadevan, A., & Tolsgaard, M. G. (2022). Does group size matter during collaborative skills learning? a randomised study. *Medical Education*, 56(6), 680–689. <https://doi.org/10.1111/medu.14791>
- Olsen, J., Aleven, V., & Rummel, N. (2017). Statistically modeling individual students’ learning over successive collaborative practice opportunities. *Journal of Educational Measurement*, 54(1), 123–138. <https://doi.org/10.1111/jedm.12137>
- Plonka, L., Segal, J., Sharp, H., & Van Der Linden, J. (2011). Collaboration in pair programming: Driving and switching. In A. Sillitti, O. Hazzan, E. Bache, & X. Albaladejo (Eds.), *Agile processes in software engineering and extreme programming* (pp. 43–59, Vol. 77). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-20677-1_4
- Salomons, N., & Scassellati, B. (2024). Time-dependant Bayesian knowledge tracing—Robots that model user skills over time. *Frontiers in Robotics and AI*, 10. <https://doi.org/10.3389/frobt.2023.1249241>
- Scott, S. D., Carpendale, M. S. T., & Inkpen, K. (2004). Territoriality in collaborative tabletop workspaces. *Proceedings of the 2004 ACM conference on Computer supported cooperative work*, 294–303. <https://doi.org/10.1145/1031607.1031655>
- Song, M. H., Park, J. A., & Park, J. (2024). Peer assessment and individual contribution analysis for measuring collaborative problem solving capability. *Current Psychology*, 43(46), 35261–35269. <https://doi.org/10.1007/s12144-024-07040-y>
- Storch, N. (2005). Collaborative writing: Product, process, and students’ reflections. *Journal of Second Language Writing*, 14(3), 153–173. <https://doi.org/10.1016/j.jslw.2005.05.002>
- Sun, C., Shute, V. J., Stewart, A. E., Beck-White, Q., Reinhardt, C. R., Zhou, G., Duran, N., & D’Mello, S. K. (2022). The relationship between collaborative problem solving behaviors and solution outcomes in a game-based learning environment. *Computers in Human Behavior*, 128, 107120. <https://doi.org/10.1016/j.chb.2021.107120>
- Volet, S., Vauras, M., Salo, A.-E., & Khosa, D. (2017). Individual contributions in student-led collaborative learning: Insights from two analytical approaches to explain the quality of group outcome. *Learning and Individual Differences*, 53, 79–92. <https://doi.org/10.1016/j.lindif.2016.11.006>
- Von Davier, A. A., & Halpin, P. F. (2013). Collaborative problem solving and the assessment of cognitive skills: Psychometric considerations. *ETS Research Report Series*, 2013(2). <https://doi.org/10.1002/j.2333-8504.2013.tb02348.x>
- Walker, E., Rummel, N., & Koedinger, K. R. (2014). Adaptive intelligent support to improve peer tutoring in algebra. *International Journal of Artificial Intelligence in Education*, 24(1), 33–61. <https://doi.org/10.1007/s40593-013-0001-9>
- Webb, N. M. (1993). Collaborative group versus individual assessment in mathematics: Processes and outcomes. *Educational Assessment*, 1(2), 131–152. https://doi.org/10.1207/s15326977ea0102_3
- Wilson, M., Gochyyev, P., & Scalise, K. (2017). Modeling data from collaborative assessments: Learning in digital interactive social networks. *Journal of Educational Measurement*, 54(1), 85–102. <https://doi.org/10.1111/jedm.12134>
- Yuan, J., Xiao, Y., & Liu, H. (2019). Assessment of collaborative problem solving based on process stream data: A new paradigm for extracting indicators and modeling dyad data. *Frontiers in Psychology*, 10. <https://doi.org/10.3389/fpsyg.2019.00369>
- Zhang, C., Ma, H., Cui, C., Yao, Y., Xu, W., Zhang, Y., & Ma, Y. (2024). Coskt: A collaborative self-supervised learning method for knowledge tracing. *IEEE Transactions on Learning Technologies*, 17, 1476–1488. <https://doi.org/10.1109/TLT.2024.3386750>